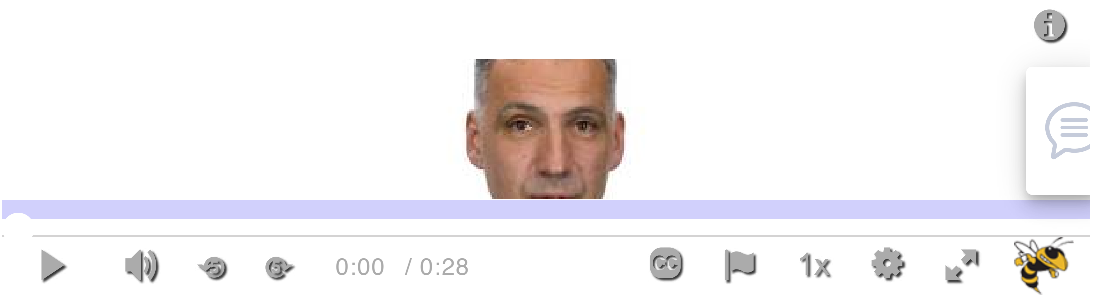
Lesson One Overview



Learning Objectives

Students will be able to:

- Define "*network science*"
- Understand its history and connections with other disciplines
- Identify some network science applications

Required and Recommended Reading

Required Reading

- Chapter-1 from A-L. Barabási, [Network Science (https://networksciencebook.com/)](https://networksciencebook.com/), 2015.

Recommended Reading

Read at least one of the following papers, depending on your interests:

- Networks in Epidemiology: **An Integrated Modeling Environment to Study the Co-evolution of Networks, Individual Behavior and Epidemics** (https://onlinelibrary.wiley.com/doi/epdf/10.1609/aimag.v31i1.2283) by Chris Barrett et al.

- Networks in Biology: **Network Inference, Analysis, and Modeling in Systems Biology (Links to an external site.)** (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2174897) by Reka Albert
- Networks in Neuroscience: **Complex brain networks: graph theoretical analysis of structural and functional systems** (http://www.nature.com/nrn/journal/v10/n3/full/nrn2575.html) by Ed Bullmore and Olaf Sporns
- Networks in Social Science: **Network Analysis in the Social Sciences** (https://www.science.org/doi/abs/10.1126/science.1165821) by Stephen Borgatti et al.
- Networks in Economics: **Economic Networks: The New Challenges** (https://www.sg.ethz.ch/publications/2009/schweitzer2009economic-networks-the) by Frank Schweitzer et al.
- Networks in Ecology: **Networks in Ecology** (https://www.sciencedirect.com/science/article/abs/pii/S1439179107000576?via%3Dihub) by Jordi Bascompte
- Networks and the Internet: **Network Topologies: Inference, Modelling and Generation** (http://www.ee.ucl.ac.uk/~mrio/papers/hamedjrnl_camera.pdf) by Hamed Haddadi et al.

Assignments

L1: Knowledge Check

L1: Quiz One